

Surface Mesh Analysis Pipeline for Al(Ga) Grain Face Characterization

Addendum: May 17, 2026 Samples

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This addendum presents curvature maps for five new grain face samples received May 17, 2026. The analysis uses the same pipeline described in the main report; only the new data and its particularities are discussed here.

Source code:

- Repository: <https://github.com/mikanystrom/intel-async> (branch `al-ga-surface-analysis`)
- Pipeline: https://github.com/mikanystrom/intel-async/tree/al-ga-surface-analysis/async-toolkit/m3utils/m3utils/Al_Ga

1 Data

The new samples consist of five grain face meshes exported from CloudCompare v2.14 in **ASCII PLY** format (the earlier samples used binary little-endian PLY). ASCII PLY support was added to the `plylib` reader for this dataset.

Each mesh has nine float properties per vertex: position (x, y, z) , normal (n_x, n_y, n_z) , and original coordinates.

File	Vertices	Faces	Normal
Face_1.ply	82,409	164,671	(0.12, -0.31, 0.94)
Face_2.ply	99,241	198,331	(-0.17, 0.23, 0.96)
Face_3.ply	169,341	338,426	(-0.12, -0.50, 0.86)
Face_4.ply	65,329	130,534	(-0.36, -0.49, 0.80)
Face_9.ply	82,441	164,763	(0.05, -0.44, 0.90)

All five faces are top-facing (z -component ≥ 0.80).

1.1 Coordinate units

The new data uses a different coordinate system from the earlier samples. Empirically, a scale factor of 100 converts the file coordinates to micrometers, yielding face extents of 1–3 mm consistent with the earlier dataset. The `plydemo` tool accepts a `-scale` parameter for this purpose:

```
plydemo -contour outdir -scale 100 data_2026_05_17/Face_1.ply
```

2 Curvature Atlas

The following pages present cotangent-Laplacian mean curvature maps for all five faces. All plots share a common color scale of $[-3, +3] \text{ mm}^{-1}$, with gray at zero (flat), red for convex, and blue for concave regions. Coordinates are in micrometers. Axis scales are identical in x and y .

Sample data_2026_05_17





